

* راح سلاقوا هون 21 سوال سنوات

* في مكان 22 سوال سنوات طليتم بشرى

صحت كل تجربة ضروري ستوفوهم لانه

اُغلبه كل وارقام.

civilian team

① A student wants to measure the volume of cylinder he used vernier Caliper to measurements the diameter which is $d = (1.50 \pm 0.05) \text{ mm}$, and used the micrometer to measure the which is $h = (2.21 \pm 0.005) \text{ mm}$. Then the volume of this cylinder in mm^3 is %

$$r = \frac{d}{2}$$

- A. 3.57 ± 0.13 B. 3.90 ± 0.05 C. 3.9 ± 0.27 D. 3.57 ± 0.005

The volume $= \pi r^2 h \Rightarrow V = \pi \times (0.75)^2 \times 2.21 = \boxed{3.90}$

② The units of density is :-

- A. a.m^3 B. m^2/s C. kg/m^3 d. kg/m

الجواب "C"

③ The Physical Laboratory experiments based on :-
a. Nature simulation b. mathematical logic
c. statistical behavior of systems d. All of choices

الجواب هو "د"

سارة محوس

④ According to Newton's second law :-

A. The direction of the velocity proportional to the net force of the object.

B. The velocity proportional to the net force of the object.

C. The "changing" in velocity proportional to the net force of the object.

D. none of choices.

الجواب هو "C"

⑤ A student measured the side length of square, and it was: $12 \text{ cm} \pm 0.02 \text{ cm}$. What ^{is} the relative error in the area of the square ($\Delta A/A$)?

- A. 3.3×10^{-3} B. 3.3×10^{-5} C. 1.7×10^{-3} d. none of the choices.

The area = $L^2 \Rightarrow \frac{\Delta A}{A} = \frac{2\Delta L}{L} = \frac{2 \times 0.02}{12} = 3.3 \times 10^{-3}$

⑥ The work done on an moving object to stop is equal:- "A" هو الجواب

- A. it's kinetic energy B. it's potential energy C. The mechanical energy

"B" هو الجواب

⑦ In Projectile motion Exp. if the mass of the object increased then:-

A. The initial velocity will be increase B. The initial velocity will be decrease
C. The initial velocity will be not change D. I can't know.

"C" هو الجواب

⑧ In the conservation of energy exp. - mass-spring system - if the energy is conserved, then the mass will be:-

A. Oscillate B. stationary C. It's doesn't matter d. None of the above

"A" هو الجواب

⑨ Units of moment of inertia is:-

- A. N B. N/cm C. kg.m^2 d. N.m "C" هو الجواب

ساره عوش

⑩ The moment of inertia of some object depends on:-

- a. torque on it b. gravitational constant C. type of substance of the object.

d. geometrical distribution of mass in the space.

"d" هو الجواب

⑪ The density of an rigid body depends on:-
A. its mass B. its volume C. Both mass and volume d. none of the choice
الجواب هو "C"

⑫ In Projectile motion EXP. the rotational motion of the ball is Occur due to:-
A. tension b. friction C. mass of the ball d. gravity
الجواب هو "ب"

⑬ The Sum of changes in Potential and Kinetic energies:-
A. Zero B. The Potential energy C. Kinetic energy d. none
الجواب هو "A"

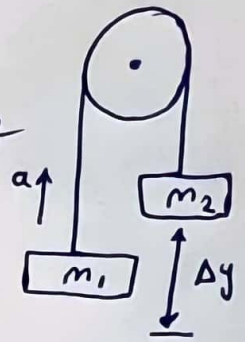
⑭ The changing in total mechanical energy of the system equals:-
a. The total Kinetic energy b. Zero C. The total Potential energy
d. The work done on the system
الجواب هو "ب"

⑮ The moment of inertia of an object depends on:-
a. The mass of the object b. The distribution of its mass on space
c. The torque on it d. (a) and (b) e. (a, b and c)
الجواب هو "د"

⑯ In mass-spring experiment, if the mechanical energy is conserved then the motion is:-
A. straight line b. stationary C. Parabola d. Oscillatory
الجواب هو "د"

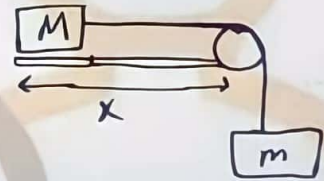
سارة
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- (17) The only external force acting on the system is:
 A. gravity B. Tension C. none d. friction
 "A" هو الجواب



- (18) A student uses the ruler to measure a diameter of some circle, if the measured value is 13mm, then the radius of this circle is:-
 a. 6.5mm b. none of the choices c. 13mm d. 7mm
 "A" هو الجواب السؤال calculated بدون نسبة مخطئي

- (19) In the Conservation of energy experiment. A student obtain the relation between the distance x and the square time as:-
 $x = 65t^2$, where x in (cm) and t in (s). if $M = 1200g$ and $m = 300g$. then the estimated value of the gravitational acceleration g in (cm/s^2) is:-
 A. 136 B. 100 C. 650 D. 300



The slope = $\frac{\Delta x}{\Delta t^2} = \frac{65t^2}{t^2} \Rightarrow$ The slope = $\frac{65t^2}{t^2}$
 $g_{exp} = \frac{2(M+m)}{m} \times \text{The slope} \Rightarrow g_{exp} = \frac{2(1200+300)}{300} \times 65$

"C" هو الجواب

- (20) In Newton's 2nd law exp. Consider the net force constant, then the acceleration of the system:
 A. will increase as total mass increases.
 B. will increase as total mass decreases.
 C. will remain constant as total mass increase.
 D. will remain constant as total mass decrease.
 "B" هو الجواب

لاره
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- ② The constant acceleration of gravity is 9.8 m/s^2 . This means that:
- A. The acceleration of the body increases by 9.8 m/s^2 during each second.
 - B. The body falls 9.8 m during the first second only.
 - C. The body falls 9.8 m during each second.
 - D. The speed of the body increases by 9.8 m/s during each second.
- الجواب هو "D"

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